

```

. /* Simulation of simple OLS */
.
. clear all
.
. set seed 12345
.
. /*****
> You can modify the following quantities
> *****/
.
. global n "150"          // no of observations

. global sd_error "1.0"    // sd of the error term

. global s_x "3.2"         // "spreadout" of x

. global a "1.5"           // intercept

. global b "2.0"           // slope

. set obs $n
number of observations (_N) was 0, now 150

. gen e = rnormal() * $sd_error

. gen x = runiform() * $s_x

. gen y = $a + $b * x + e
.
. sum x y

```

Variable	Obs	Mean	Std. Dev.	Min	Max
x	150	1.516282	.9667103	.0181646	3.192279
y	150	4.62719	2.231906	-.7227519	10.90633

```

. regress y x

```

Source	SS	df	MS	Number of obs	=	150
Model	588.958086	1	588.958086	F(1, 148)	=	568.70
Residual	153.271141	148	1.03561582	Prob > F	=	0.0000
Total	742.229227	149	4.98140421	R-squared	=	0.7935
				Adj R-squared	=	0.7921
				Root MSE	=	1.0177

y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
x	2.056614	.0862402	23.85	0.000	1.886192 2.227035
_cons	1.508783	.1549305	9.74	0.000	1.202622 1.814945

```

.
. /*****
> Usually we would repeat the above process many times
> to call it "Monte Carlo simulations." But we only
> did it once above because we didn't want to get into
> more advanced Stata programming and other issues.
> *****/
.
. // (1) What if you move the x's away from 0 to see the
. // impact on Var(a)?
.
. gen xnew = x+100

. regress y xnew

```

Source	SS	df	MS	Number of obs	=	150
				F(1, 148)	=	568.70
Model	588.95788	1	588.95788	Prob > F	=	0.0000
Residual	153.271347	148	1.03561721	R-squared	=	0.7935
				Adj R-squared	=	0.7921
Total	742.229227	149	4.98140421	Root MSE	=	1.0177

y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
xnew	2.056614	.0862403	23.85	0.000	1.886193 2.227035
_cons	-204.1526	8.75519	-23.32	0.000	-221.454 -186.8513

```

.
. // (2) What if we multiply y by 100?
.
. gen y100 = y*100

. regress y100 x

```

Source	SS	df	MS	Number of obs	=	150
				F(1, 148)	=	568.70
Model	5889580.83	1	5889580.83	Prob > F	=	0.0000
Residual	1532711.43	148	10356.1583	R-squared	=	0.7935
				Adj R-squared	=	0.7921
Total	7422292.26	149	49814.042	Root MSE	=	101.77

y100	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
x	205.6614	8.624022	23.85	0.000	188.6192 222.7035
_cons	150.8783	15.49305	9.74	0.000	120.2621 181.4945

```

.
. // (3) What if we multiply x by 100?
.
. gen x100 = x*100

. regress y x100

```

Source	SS	df	MS	Number of obs	=	150
Model	588.958087	1	588.958087	F(1, 148)	=	568.70
Residual	153.27114	148	1.03561581	Prob > F	=	0.0000
Total	742.229227	149	4.98140421	R-squared	=	0.7935
				Adj R-squared	=	0.7921
				Root MSE	=	1.0177

y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
x100	.0205661	.0008624	23.85	0.000	.0188619 .0222703
_cons	1.508783	.1549305	9.74	0.000	1.202621 1.814945

```

.
. // (4) What if we multiply both by 100?
.
. regress y100 x100

```

Source	SS	df	MS	Number of obs	=	150
Model	5889580.84	1	5889580.84	F(1, 148)	=	568.70
Residual	1532711.42	148	10356.1582	Prob > F	=	0.0000
Total	7422292.26	149	49814.042	R-squared	=	0.7935
				Adj R-squared	=	0.7921
				Root MSE	=	101.77

y100	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
x100	2.056614	.0862402	23.85	0.000	1.886192 2.227035
_cons	150.8783	15.49305	9.74	0.000	120.2621 181.4945